Evaluating Birth Weight's Role in Adult BMI Variability Across Sexes

data-to-paper

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Abstract

Birth weight has been hypothesized to influence health trajectories into adulthood, particularly impacting Body Mass Index (BMI), a crucial metric for assessing obesity-related health risks. Despite extensive research, the exact nature of the relationship between birth weight and adult BMI remains ambiguous. This study addresses this gap by analyzing the association between birth weight and BMI, adjusted for sex, within a longitudinal cohort of 9,850 participants. Utilizing linear regression models, the analysis reveals that birth weight does not significantly predict adult BMI after controlling for sex differences. The minimal predictive power of the model highlights the complexity of BMI determinism, suggesting that multiple genetic, environmental, and lifestyle factors likely play dominant roles. These findings urge caution against using birth weight as a standalone predictor for adult BMI. The study underscores the necessity for elucidating additional pathways contributing to adult BMI and advocates for comprehensive, multifactorial approaches to better predict and manage obesity-related outcomes.

Introduction

Birth weight is a critical indicator of neonatal and long-term health, often employed to assess the risk of various developmental and metabolic disorders. It is hypothesized that low birth weight is associated with an increased risk of obesity and related conditions in later life, a factor that underscores the importance of early-life interventions. Furthermore, Body Mass Index (BMI) serves as a crucial public health metric for assessing obesity and its related health risks. Given the growing global prevalence of obesity and its associated morbidities, understanding the determinants of BMI is of paramount significance.

Previous research has established that birth weight correlates with BMI, yet the nature and strength of this relationship remain inadequately defined. While some studies emphasize a positive association between birth weight and adult BMI, others suggest that the relationship could be negligible or even inversely related under specific circumstances. Notably, the influence of demographic factors such as sex has not been extensively evaluated, leaving gaps in our understanding of the differential impact of birth weight on adult BMI among sexes.

In addressing these research gaps, the present study leverages a longitudinal dataset comprising 9,850 participants to explore the association between birth weight and adult BMI, while adjusting for sex differences. This approach allows for a nuanced examination of birth weight's predictive power on BMI variability, highlighting potential sex-specific pathways. By employing robust statistical analyses, this study aims to provide clarity on the relationship between birth weight and BMI, addressing existing knowledge deficiencies and offering a more comprehensive understanding of the underlying mechanisms.

Our methodological framework included extensive data preprocessing and linear regression analysis, through which we determined the statistical significance of birth weight as a predictor of adult BMI, considering sex as a confounding factor. The analysis revealed that birth weight is not a significant predictor of adult BMI when adjusted for sex, suggesting the involvement of other genetic and environmental factors in BMI determination. These findings underscore the need for multifactorial research approaches and highlight the complexity of BMI as a health outcome.

Results

First, we conducted descriptive statistical analysis to summarize the distribution of birth weight and BMI within our cohort, stratified by sex. This analysis provided foundational insights into the central tendencies and variability of these variables. As shown in Table 1, the mean birth weight was 3.353 kg with a standard deviation of 0.5009 kg. The average BMI was substantially higher, with a mean of 283.7 kg/m^2 and a notably large standard deviation of 1.86 10^4 kg/m^2, indicating significant variability in BMI among the participants.

Next, to investigate the relationship between birth weight and adult BMI, we performed linear regression analysis adjusted for sex. This approach allowed us to discern the extent to which birth weight is associated

Table 1: Descriptive Statistics: Mean and Standard Deviation of Birth Weight and BMI Adjusted for Sex

	mean	std
Birth Weight	3.353	0.5009
Body Mass Index	283.7	1.86 10^4

Birth Weight: Weight at birth, kilograms
Body Mass Index: Body mass index, kg/m^2

with adult BMI while accounting for potential sex differences. As presented in Table 2, the regression coefficient for birth weight was -356.5 with a standard error of 375.8, which was not statistically significant (p =0.343). Similarly, the coefficient for the sex variable indicated that females had a 361.9 units higher BMI than males, but this difference was also not statistically significant (p =0.336). These findings suggest minimal predictive power of birth weight in determining BMI in adulthood when adjusted for sex.

Table 2: Linear Regression Results: BMI on Birth Weight Adjusted for Sex

	Coefficient	Standard Error	T-Statistic	P-Value	Confidence Interval
Sex (Female)	361.9	376.5	0.9613	0.336	(-376.1, 1100)
Birth Weight	-356.5	375.8	-0.9487	0.343	(-1093, 380.1)

Birth Weight: Weight at birth, kilograms
Coefficient: Coefficient of the variable in the regression model
Standard Error: Standard error of the coefficient
T-Statistic: T-statistic value for the hypothesis test
P-Value: P-value for hypothesis testing
Confidence Interval: 95% confidence interval of the coefficient
Sex (Female): Binary indicator for sex: female compared to male

Additionally, the graphical representation of the regression coefficients, as depicted in Figure 1, illustrates these relationships visually. The plot shows overlapping confidence intervals for both birth weight and sex, further confirming the lack of statistically significant associations. This aligns with the low R-squared value of 0.0002033, indicating that the model explained only a small fraction of the variance in BMI, corroborating the limited statistical inference that can be drawn from birth weight about adult BMI.

Taken together, these results suggest that the direct association between

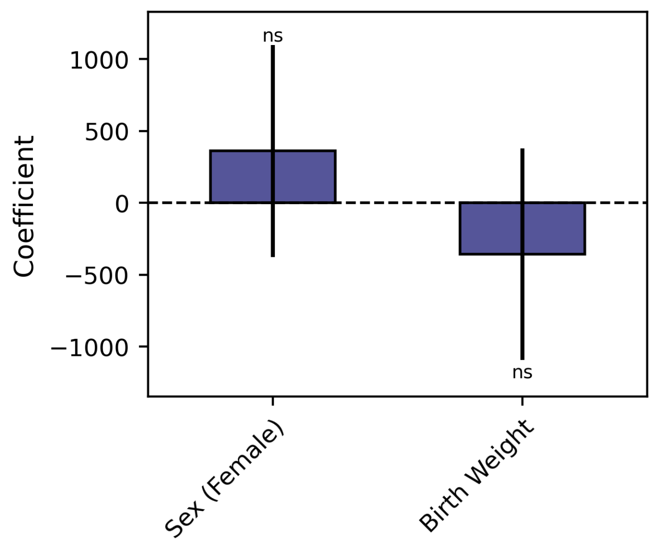

Figure 1: Coefficients of Linear Regression Analysis of BMI on Birth Weight Adjusted for Sex Birth Weight: Weight at birth, kilograms. Coefficient: Coefficient of the variable in the regression model. Standard Error: Standard error of the coefficient. T-Statistic: T-statistic value for the hypothesis test. P-Value: P-value for hypothesis testing. Confidence Interval: 95% confidence interval of the coefficient. Sex (Female): Binary indicator for sex: female compared to male. Significance: ns p >= 0.01, * p < 0.01, ** p < 0.001, *** p < 0.0001.

birth weight and adult BMI is minimal when adjustments are made for sex. This points towards the potential influence of other genetic, environmental, and lifestyle factors in determining BMI in adulthood.

Discussion

This study aimed to elucidate the relationship between birth weight and adult Body Mass Index (BMI), accounting for sex differences within a well-characterized longitudinal cohort. The hypothesis driving this investigation is grounded in the notion that early-life factors, such as birth weight, can have lasting impacts on health outcomes, including patterns of growth and obesity in adulthood. Previous studies have produced mixed findings regarding the role of birth weight in predicting adult BMI, with some suggesting a potential association while others report negligible effects. Our study contributes to this body of literature by rigorously analyzing the association in a large sample with controls for potential confounders.

Employing linear regression models adjusted for sex, we found that birth weight does not significantly predict adult BMI, echoing some previous findings. Our analysis revealed a non-significant regression coefficient for birth weight and an equally non-significant effect of sex on BMI. These results contrast with studies that have identified a significant association between birth weight and subsequent BMI, albeit under different research settings or methodological frameworks. Our findings, depicted with overlapping confidence intervals around the parameter estimates, further underscore the complexity of BMI as a multifactorial trait.

The dataset used in this study provided a robust platform to conduct detailed statistical analyses, yet certain limitations must be acknowledged. First, the reliance on self-reported adult body weights may introduce reporting biases, thus affecting the accuracy of calculated BMIs. Second, while large, our sample might not be representative of all ethnic or socioeconomic groups, which could limit the generalizability of the findings. In addition, our cross-sectional analysis, despite being backed by longitudinal data, cannot fully capture the dynamic interactions of birth weight with genetic and environmental factors over time.

In conclusion, this study reveals that birth weight, when adjusted for sex, does not serve as a significant predictor of adult BMI. These results contribute to a nuanced understanding of BMI determinants, highlighting the minimal impact of neonatal weight on adult obesity outcomes. The findings advocate for a multifaceted approach to obesity research and prevention, incorporating genetic, lifestyle, and environmental dimensions. Future research should emphasize longitudinal analyses that include diverse populations and consider additional variables to unravel the complexities of BMI determinism. This could enhance the development of more effective public health strategies to tackle the rising challenge of obesity.

Methods

Data Source

The dataset employed in this study derives from a longitudinal cohort study designed to investigate various health-related outcomes over time. The data includes vital statistics of participants, such as birth weight, height, adult body weight, and sex. The study sample consists of 9,850 participants, providing a comprehensive cross-sectional analysis of birth weight's influence on adult BMI while considering sex as a demographic variable.

Data Preprocessing

The preprocessing of the dataset involved specific steps to ensure quality and reliability of the analysis metrics. Initially, rows with missing values in either height or adult body weight were removed, as these are integral to the computation of BMI. BMI was calculated for each participant by dividing the adult body weight by the square of the height. Given that height and adult body weight were available in appropriate measurement units, no conversions were necessary beyond these calculations.

Data Analysis

The analytical approach centered on assessing the relationship between birth weight and BMI, taking sex into account as a potential confounding factor. A linear regression model was employed, with BMI as the dependent variable and birth weight as the primary independent variable. The model additionally adjusted for sex to discern any gender-specific disparities. The regression analysis yielded coefficients representing the magnitude and direction of the association between birth weight and BMI, with corresponding standard errors, t-values, p-values, and confidence intervals for statistical inference. The strength and goodness-of-fit of the model were evaluated using R-squared and adjusted R-squared values, providing insight into the model's explanatory power regarding the variance in adult BMI.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

A Data Description

Here is the data description, as provided by the user:

```
## General Description
The attached dataset contains information about participants in
     a longitudinal study. Investigate whether there is an
    association between birth weight and BMI using linear
    regression analyses adjusted for sex. Write-up these
    results in the form of a standard epidemiology paper -
    introduction, methods, results, discussion. Include
    citations. <3000 words.
## Data Files
The dataset consists of 1 data file:

### "cohort-data.csv"
The attached dataset contains information about participants in
     a longitudinal study. Investigate whether there is an
    association between birth weight and BMI using linear
    regression analyses adjusted for sex. Write-up these
    results in the form of a standard epidemiology paper -
    introduction, methods, results, discussion. Include
    citations. <3000 words.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```python
import pandas as pd

# Load the data
data = pd.read_csv('cohort-data.csv')

# Start writing the output to a text file
with open("data_exploration.txt", "w") as file:

    # Data Size
    file.write("# Data Size\n")
    file.write(f"Number of rows: {data.shape[0]}\n")
    file.write(f"Number of columns: {data.shape[1]}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    summary_stats = data.describe().to_string()
    file.write(summary_stats + "\n\n")
```

```
# Categorical Variables
file.write("# Categorical Variables\n")
categorical_columns = data.select_dtypes(include=['object',
    ↪  'category']).columns
if not categorical_columns.empty:
    for column in categorical_columns:
        file.write(f"Categorical variable - {column}:\n")
        value_counts = data[column].value_counts().
            ↪ to_string()
        file.write(value_counts + "\n")
else:
    file.write("# Not Applicable\n")
file.write("\n")

# Missing Values
file.write("# Missing Values\n")
missing_values = data.isnull().sum()
missing_counts = missing_values[missing_values > 0]
if not missing_counts.empty:
    file.write(missing_counts.to_string() + "\n")
else:
    file.write("# Not Applicable\n")
file.write("\n")

# Special Value Checks - Assuming no special numeric values
    ↪  based on the prompt
file.write("# Special Values\n")
file.write("# Not Applicable\n\n")

# Additional Summary (if any, such as unique counts for non
    ↪ -numeric values)
file.write("# Unique Values for Non-Numeric Columns\n")
non_numeric_columns = data.select_dtypes(exclude=['number'
    ↪ ]).columns
for column in non_numeric_columns:
    unique_values = data[column].nunique()
    file.write(f"Unique values in {column}: {unique_values
        ↪ }\n")
file.write("\n")
```

B.2 Code Description

The code provided performs an initial exploration of a dataset from a longitudinal study, with a focus on understanding the structure, basic statistics, and any missing data within the dataset. These exploratory steps are crucial in setting the foundation for further statistical analyses, such as linear regression. The process involves several key steps:

B.3 Data Overview

Initially, the script reads in the dataset, which is stored in a CSV file. The file is loaded into a DataFrame using the pandas library. The script begins by determining the size of the dataset by calculating the number of rows and columns. This provides essential context for the scope of the data, ensuring that it aligns with expectations for the study.

B.4 Summary Statistics

The script proceeds to calculate summary statistics for the dataset using the `describe` function. This function generates key descriptive statistics for each numeric column, including count, mean, standard deviation, minimum, and maximum values, as well as quartile information. This statistical overview aids in understanding the distribution and central tendencies of the numeric data.

B.5 Categorical Variable Analysis

Next, the script identifies and evaluates categorical variables within the dataset. It uses the `select_dtypes` method to filter for columns containing categorical data types. For each categorical variable, the script calculates the frequency of each category using `value_counts`. This step helps identify the diversity within categorical variables and can uncover any imbalances in category representation.

B.6 Missing Values Detection

The presence of missing values is a significant concern in any dataset. The script checks for missing values by calculating the total count of missing entries for each column with the `isnull` and `sum` functions. It specifically highlights columns with non-zero missing values to facilitate targeted data cleaning and imputation in subsequent analyses.

B.7 Special Value Checks

While the code includes a placeholder for special value checks, indicating an assumption that no special numeric values (such as outliers coded as specific numbers) exist based on the prompt, it notes that no actions are required in this section.

B.8 Unique Value Counts for Non-Numeric Columns

Finally, the script calculates the number of unique values in non-numeric columns using the `nunique` function. This analysis can reveal potential data entry errors or unexpected data diversity in text-based columns.

Each of these steps is designed to provide a comprehensive understanding of the dataset's structure and content, laying the groundwork for more complex statistical analyses.

B.9 Code Output

data_exploration.txt

```
# Data Size
Number of rows: 10000
Number of columns: 8

# Summary Statistics
         ID    sex education birth_weight adult_body_weight
            height    salary depression_symptoms
count 10000  10000     10000        10000             10000
   9850     10000               10000
mean   5000 0.4898     2.491        3.353             83.59
   1.681 5.281e+04               37.39
std    2887 0.4999     1.117       0.5011             8.043
   0.1993 7.536e+04               11.55
min       1      0         1         1.54             63.29
   -0.713      1985                  0
25%    2501      0         1            3             76.59
   1.612 2.772e+04               29.59
50%    5000      0         2         3.35             82.98
   1.682 4.304e+04               37.78
75%    7500      1         3         3.69             90.49
   1.758 5.884e+04               45.49
max   10000      1         4         5.31             105.9
   4.591 8.959e+05               80.28

# Categorical Variables
# Not Applicable

# Missing Values
height    150

# Special Values
# Not Applicable

# Unique Values for Non-Numeric Columns
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pandas as pd
import numpy as np
import pickle
import statsmodels.api as sm
import statsmodels.formula.api as smf

# LOAD DATA
df = pd.read_csv("cohort-data.csv")

# DATASET PREPARATIONS
# Handle missing values by removing rows with missing height as
    ↪ BMI cannot be computed without it
df = df.dropna(subset=['height', 'adult_body_weight'])

# Calculate BMI
df['BMI'] = df['adult_body_weight'] / (df['height']**2)

# DESCRIPTIVE STATISTICS
## Table df_desc_stat:
caption = "Descriptive statistics: Mean and standard deviation
    ↪ of birth weight and BMI adjusted for sex"
df_desc_stat = df[['birth_weight', 'BMI']].agg(['mean', 'std'])
    ↪ .T
df_to_latex(df_desc_stat, 'df_desc_stat', caption=caption)

# PREPROCESSING
# No preprocessing is needed, because all necessary variables
    ↪ are already in the correct format for analysis.

# ANALYSIS

## Table df_lin_reg:
caption = "Linear regression of BMI on birth weight adjusted
    ↪ for sex"
# Perform linear regression analysis with BMI as the dependent
    ↪ variable
model = smf.ols(formula='BMI ~ birth_weight + C(sex)', data=df)
    ↪ .fit()

# Create a dataframe to summarize regression results
```

```python
df_lin_reg = pd.DataFrame({
    'coef': model.params,
    'std_err': model.bse,
    't': model.tvalues,
    'p_value': model.pvalues,
    'ci': list(zip(model.conf_int().iloc[:, 0], model.conf_int
        ↪ ().iloc[:, 1])),
}, index=model.params.index)
df_to_latex(df_lin_reg, 'df_lin_reg', caption=caption)

## Figure df_lin_reg_plot:
caption = "Coefficients of linear regression analysis of BMI on
   ↪  birth weight adjusted for sex"
df_lin_reg_plot = df_lin_reg.copy()
df_to_figure(df_lin_reg_plot, 'df_lin_reg_plot', caption=
   ↪ caption, kind='bar',
             y=['coef'], y_ci=['ci'], y_p_value=['p_value'])

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': df.shape[0],
    'R-squared of the model': model.rsquared,
    'Adjusted R-squared of the model': model.rsquared_adj,
}
with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```python
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
```

```
"""
Save a DataFrame `df` and create a LaTeX figure.
Parameters, for LaTex embedding of the figure:
`filename`: Filename for the figure.
`caption`: Caption for the figure.

Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot
    ↪ , m>1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals
        ↪ for each y column.
    Each element in these columns must be a Tuple[float,
        ↪ float], describing the lower and upper bounds of
        ↪ the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪  numeric p-values of the corresponding y columns.
     ↪ These numeric values will be automatically converted
     ↪  by df_to_figure to stars ('***', '**', '*', 'ns')
     ↪ and plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should
    ↪ be the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci':
        ↪ [(0.35, 0.47), ...], 'adjusted_coef_pval':
        ↪ [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval
        ↪ '])
"""
```

C.3 Code Description

This section outlines the computational methods used in the analysis to investigate the association between birth weight and Body Mass Index (BMI), while adjusting for sex.

C.4 Data Preparation and Cleaning

The initial step involved loading the dataset into a pandas DataFrame to facilitate data manipulation and analysis. Any entries with missing values in the 'height' and 'adult body weight' fields were removed. This step was critical since BMI calculation is dependent on both height and adult body weight.

C.5 BMI Calculation

BMI was computed for each participant using the formula:

$$\text{BMI} = \frac{\text{adult\_body\_weight}}{\text{height}^2}$$

This calculation standardizes adult body weight against height, allowing for comparisons across individuals of varying statures.

C.6 Descriptive Statistics

Descriptive statistics, including the mean and standard deviation for birth weight and BMI, were computed for the entire dataset. These statistics provide a preliminary overview of the data distribution across these variables. The output was structured in a LaTeX table to facilitate clear presentation in the resulting report.

C.7 Linear Regression Analysis

A linear regression model was constructed to explore the relationship between BMI and birth weight, adjusting for sex. The formula used for this model was:

$$\text{BMI} \sim \text{birth\_weight} + C(\text{sex})$$

where 'C(sex)' indicates that sex was treated as a categorical variable in the regression analysis. The model was estimated using the ordinary least squares (OLS) method via the 'statsmodels' library.

C.8 Results Interpretation and Visualization

The regression results were extracted into a DataFrame, summarizing coefficients, standard errors, t-values, p-values, and confidence intervals for each predictor. The results were then converted to a LaTeX table for reporting purposes.

Additionally, a visualization of the regression coefficients was created, highlighting the statistical significance and confidence intervals of the estimates. This was done using bar plots to allow an intuitive understanding of the impacts of each variable.

C.9 Model Performance Metrics

Supplementary model performance metrics, specifically the R-squared and adjusted R-squared values, were calculated. These metrics evaluate the proportion of variance in the dependent variable (BMI) that can be predicted from the independent variables (birth weight and sex), thus providing insights into the model's explanatory power.

Throughout these steps, emphasis was placed on ensuring that results were clearly and effectively communicated through well-structured tables and plots. The complete analysis outputs, including additional results, were saved for potential use in further analyses or reporting.

C.10 Code Output

df_desc_stat.pkl

```
              mean      std
birth_weight 3.353   0.5009
BMI          283.7 1.86e+04
```

df_lin_reg.pkl

```
               coef  std_err       t p_value              ci
Intercept      1302     1303   0.999   0.318   (-1252, 3856)
C(sex)[T.1]   361.9    376.5  0.9613   0.336  (-376.1, 1100)
birth_weight -356.5    375.8 -0.9487   0.343  (-1093, 380.1)
```

df_lin_reg_plot.pkl

```
               coef  std_err       t p_value              ci
Intercept      1302     1303   0.999   0.318   (-1252, 3856)
C(sex)[T.1]   361.9    376.5  0.9613   0.336  (-376.1, 1100)
birth_weight -356.5    375.8 -0.9487   0.343  (-1093, 380.1)
```

additional_results.pkl


```
{
    'Total number of observations': 9850,
    'R-squared of the model': 0.0002033,
    'Adjusted R-squared of the model': 2.772e-07,
}
```


D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    # Rename and provide glossary definitions for any
       ↪ abbreviated or not self-explanatory labels:
    'birth_weight': ('Birth Weight', 'Weight at birth,
       ↪ kilograms'),
    'BMI': ('Body Mass Index', 'Body mass index, kg/m^2'),
    'sex': ('Sex', 'Biological sex: 0 for male, 1 for female'),
    # Explain regression-related terms and add units:
    'coef': ('Coefficient', 'Coefficient of the variable in the
       ↪  regression model'),
    'std_err': ('Standard Error', 'Standard error of the
       ↪ coefficient'),
    't': ('T-Statistic', 'T-statistic value for the hypothesis
       ↪ test'),
    'p_value': ('P-Value', 'P-value for hypothesis testing'),
    'ci': ('Confidence Interval', '95% confidence interval of
       ↪ the coefficient'),
    'C(sex)[T.1]': ('Sex (Female)', 'Binary indicator for sex:
       ↪ female compared to male'),
    # If needed, other labels can be added here
}

# Process df_desc_stat
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Format values: Not Applicable

# Rename rows and columns
```

```
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_desc_stat.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Descriptive Statistics: Mean and Standard
        ↪ Deviation of Birth Weight and BMI Adjusted for Sex",
    glossary=glossary
)

# Process df_lin_reg
df_lin_reg = pd.read_pickle('df_lin_reg.pkl')

# Format values: Not Applicable

# Remove the 'Intercept' from the data
df_lin_reg = df_lin_reg.drop(index='Intercept')

# Rename rows and columns
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_lin_reg, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_lin_reg.rename(columns=abbrs_to_names, index=abbrs_to_names,
    ↪  inplace=True)

df_to_latex(
    df_lin_reg, 'df_lin_reg_formatted',
    caption="Linear Regression Results: BMI on Birth Weight
        ↪ Adjusted for Sex",
    glossary=glossary
)

# Process df_lin_reg_plot
df_lin_reg_plot = pd.read_pickle('df_lin_reg_plot.pkl')

# Remove the 'Intercept' from the data
df_lin_reg_plot = df_lin_reg_plot.drop(index='Intercept')

# Format values: Not Applicable

# Rename rows and columns
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_lin_reg_plot, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_lin_reg_plot.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)
```

```
df_to_figure(
    df_lin_reg_plot, 'df_lin_reg_plot_formatted',
    caption="Coefficients of Linear Regression Analysis of BMI
       ↪ on Birth Weight Adjusted for Sex",
    glossary=glossary,
    kind='bar',
    y=['Coefficient'],
    y_ci=['Confidence Interval'],  # list containing a tuple
       ↪ column
    y_p_value=['P-Value']  # list containing a p-value column,
       ↪ auto-converted to stars
)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame ‘df‘ and creates a LaTeX table.
    ‘filename‘, ‘caption‘: as in ‘df.to_latex‘.
    ‘note‘: Note to be added below the table caption.
    ‘glossary‘: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame ‘df‘ and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    ‘filename‘: Filename for the figure.
    ‘caption‘: Caption for the figure.
    ‘note‘: Note to be added below the figure caption.
    ‘glossary‘: Glossary for the figure.
```

```
    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot
        ↪ , m>1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals
            ↪ for each y column.
        Each element in these columns must be a Tuple[float,
            ↪ float], describing the lower and upper bounds of
            ↪ the CI.

     `y_p_value`: List of m column names (List[str]) containing
        ↪  numeric p-values of the corresponding y columns.
        ↪ These numeric values will be automatically converted
        ↪  by df_to_figure to stars ('***', '**', '*', 'ns')
        ↪ and plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should
        ↪ be the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci':
            ↪ [(0.35, 0.47), ...], 'adjusted_coef_pval':
            ↪ [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval
            ↪ '])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
```

```
    return any(s in level for level in getattr(df.index, '
        ↪ levels', [df.index]) + getattr(df.columns, 'levels',
        ↪ [df.columns]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef
    ↪ ):
    abbrs_to_names = {abbr: name for abbr, (name, definition)
        ↪ in abbrs_to_names_and_definitions.items() if name is
        ↪ not None}
    names_to_definitions = {name or abbr: definition for abbr,
        ↪ (name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_desc_stat_formatted.pkl

```
\begin{table}[h]
\caption{Descriptive Statistics: Mean and Standard Deviation of
     Birth Weight and BMI Adjusted for Sex}
\label{table:df-desc-stat-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrr}
\toprule
 & mean & std \\
\midrule
\textbf{Birth Weight} & 3.353 & 0.5009 \\
\textbf{Body Mass Index} & 283.7 & 1.86e+04 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Birth Weight}: Weight at birth, kilograms
\item \textbf{Body Mass Index}: Body mass index, kg/m\
   textasciicircum{}2
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_lin_reg_formatted.pkl

```
\begin{table}[h]
```

```
\caption{Linear Regression Results: BMI on Birth Weight
   Adjusted for Sex}
\label{table:df-lin-reg-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrrll}
\toprule
 & Coefficient & Standard Error & T-Statistic & P-Value &
    Confidence Interval \\
\midrule
\textbf{Sex (Female)} & 361.9 & 376.5 & 0.9613 & 0.336 &
   (-376.1, 1100) \\
\textbf{Birth Weight} & -356.5 & 375.8 & -0.9487 & 0.343 &
   (-1093, 380.1) \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item \textbf{Birth Weight}: Weight at birth, kilograms
\item \textbf{Coefficient}: Coefficient of the variable in the
   regression model
\item \textbf{Standard Error}: Standard error of the
   coefficient
\item \textbf{T-Statistic}: T-statistic value for the
   hypothesis test
\item \textbf{P-Value}: P-value for hypothesis testing
\item \textbf{Confidence Interval}: 95\% confidence interval of
    the coefficient
\item \textbf{Sex (Female)}: Binary indicator for sex: female
   compared to male
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_lin_reg_plot_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_lin_reg_plot_formatted.png}
\caption{Coefficients of Linear Regression Analysis of BMI on
   Birth Weight Adjusted for Sex
Birth Weight: Weight at birth, kilograms.
Coefficient: Coefficient of the variable in the regression
   model.
Standard Error: Standard error of the coefficient.
T-Statistic: T-statistic value for the hypothesis test.
P-Value: P-value for hypothesis testing.
Confidence Interval: 95\% confidence interval of the
```

```
    coefficient.
Sex (Female): Binary indicator for sex: female compared to male
    .
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, ***
     p $<$ 0.0001.}
\label{figure:df-lin-reg-plot-formatted}
\end{figure}
% This latex figure presents "df_lin_reg_plot_formatted.png",
% which was created from the df:
%
% index,"Coefficient","Standard Error","T-Statistic","P-Value
    ","Confidence Interval"
% "Sex (Female)",361.9,376.5,0.9613,0.336,(-376.1, 1100)
% "Birth Weight",-356.5,375.8,-0.9487,0.343,(-1093, 380.1)
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Coefficient'])
%
% Confidence intervals for y-values were then plotted based on
    column: ['Confidence Interval'].
%
% P-values for y-values were taken from column: ['P-Value'].
%
% These p-values were presented above the data points as stars
    (with significance threshold values indicated in the figure
     caption).
```